

RSIMACD Divergence Indicator & Strategy

Indicator Overview

The indicator identifies **regular** and **hidden** divergences between price and the MACD histogram, using

- **Fast MA** SMA of period *fast* (default 12)
- **Slow MA** SMA of period *slow* (default 26)
- **Signal** EMA of the MACD line with *smooth* periods (default 9)
- **Histogram** MACD Signal

Fractal Detection

Fivebar pivot highs/lows on the histogram are used as anchor points for divergences.

- **Top fractal** bar 2 is higher than bars 0, 1, 3, 4
- **Bottom fractal** bar 2 is lower than bars 0, 1, 3, 4

Divergence Logic

When a new top or bottom fractal appears, the indicator compares price highs/lows against the previous fractal

Divergence Type	Price Trend	Histogram Trend	Interpretation
Regular Bearish	Higher high	Lower high	Bearish reversal
Hidden Bearish	Lower high	Higher high	Bearish continuation
Regular Bullish	Lower low	Higher low	Bullish reversal
Hidden Bullish	Higher low	Lower low	Bullish continuation

Output Columns

The DataFrame returned by `detect_divergences()` contains:

- `macd`, `signal`, `hist`
- `fractal_top`, `fractal_bot`
- `regular_bearish`, `hidden_bearish`, `regular_bullish`, `hidden_bullish`

Trading Strategy

The corresponding backtesting strategy trades on the divergence signals:

1. ****Long entry**** Regular Bullish (or, if `--hidden`` flag is set, Hidden Bullish) divergence.
2. ****Short entry**** Regular Bearish (or, if `--hidden`` flag is set, Hidden Bearish) divergence.
3. ****Exit conditions****
 - Opposite signal appears.
 - TakeProfit (TP) hit (`--tp``).
 - StopLoss (SL) hit (`--sl``).
 - Or at the end of the data period.

Parameters

```
...  
  
--symbol    Currency pair or ticker (default: EURUSD)  
--timeframe Chart timeframe (e.g., H1, H4, D1)  
--start     Start date (YYYYMMDD)  
--end       End date (YYYYMMDD, optional)  
--fast      Fast MA period (default 12)  
--slow      Slow MA period (default 26)  
--smooth    Signal EMA period (default 9)  
--hidden    Include hidden divergences (long/short)  
--tp        TakeProfit percentage (e.g., 0.02 = 2%)  
--sl        StopLoss percentage (e.g., 0.01 = 1%)  
--size      Position size multiplier  
--plot-equity Plot equity curve after backtest  
...
```

Usage

```
```bash
```

## Plot divergences

```
python main.py plot --symbol EURUSD --timeframe H1 --start 2024-01-01
```

## Run backtest with hidden divergences and TP/SL

```
python main.py backtest --symbol GBPUSD --timeframe H1 \
 --start 2024-01-01 --tp 0.02 --sl 0.01 --hidden --plot-equity
```

```
...
```

## Statistics

The backtest prints:

- Total return, max drawdown, Sharpe ratio, profit factor, win rate, number of trades, gross profit/loss.
- Detailed trade list.
- Optional equity curve plot.

## Dependencies

- ``pandas``, ``numpy``, ``matplotlib``
- Optional ``MetaTrader5`` provider; falls back to ``yfinance``.

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For further details refer to the repository README and source code.